Uncertainty-aware CALPHAD–machine-learning screening of heat-treatment intervals

Six-solute Ni-base superalloys · open MatCalc thermodynamic database · validated against six commercial alloys

WORKFLOW

1 Two linked CALPHAD calculations

Ni–Co–Cr–Al–Mo–Ti–C composition space
480-alloy Latin-hypercube sample
matrix melting restricted to liquid + fcc
solid state — γ′, MC, $M_{23}C_6$, σ, bcc

2 Uncertainty-aware surrogates

Gaussian process, polynomial chaos,
neural-network and PCA-GP models
with calibrated prediction intervals

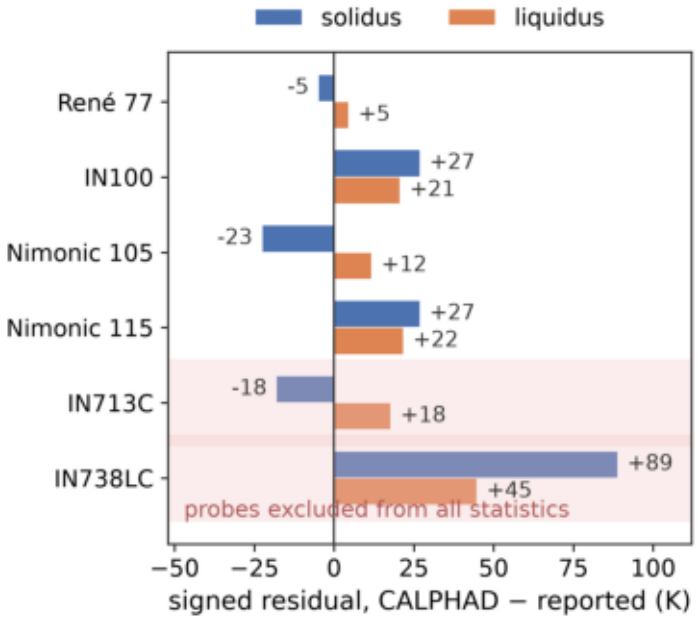
3 Validation on six commercial alloys

mean absolute errors of 20.3 K (solidus)
and 14.7 K (liquidus)
solvus cross-check: −0.1 K / +29.6 K
domain boundary chemical, not geometric

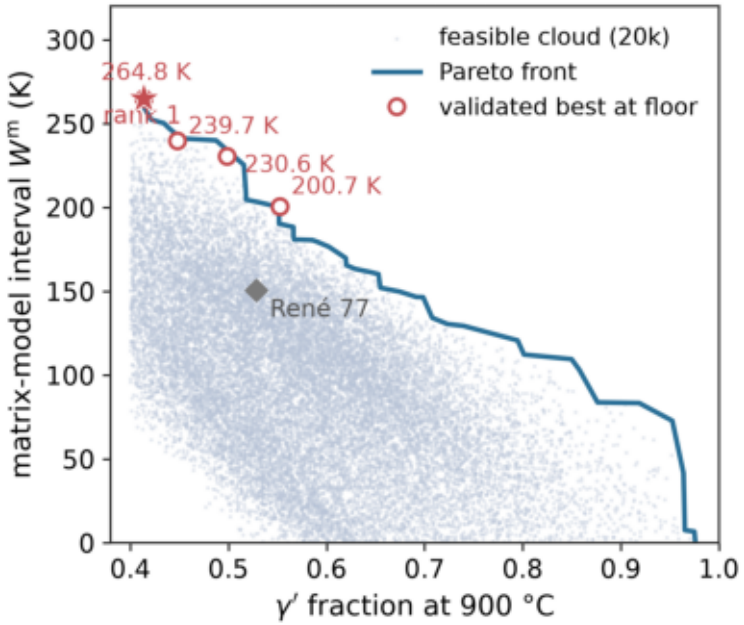
4 Inverse design and re-validation

120,000-candidate Sobol screen
Pareto front with γ′-floor sensitivity
σ-phase service screen at 750 and 900 °C

(a) Validation against calorimetry



(b) Pareto design curve



✓ Screened interval of 264.8 K — 114.3 K wider than René 77 via γ′-solvus depression

✓ The sign of the gain is preserved with 98.1 % probability across all evaluated models

! Calculated screening quantity, not a measured window — paired-DSC test pre-registered